Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/03/2004
PI Name:	Holstein, Gay R Ph.D.		
Project Title:	Anatomical Studies of Central Vestibular Adaptation-Neurolab Completion Experiment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
PI Email:	gay.holstein@mssm.edu	**Fax:**	FY 212-987-3937
PI Organization Type:	NON-PROFIT	**Phone:**	212-241-7072
Organization Name:	Mount Sinai School of Medicine		
PI Address 1:	Box 1140; Department of Neurology		
PI Address 2:	One Gustave Levy Place		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	04/01/2000	**End Date:**	03/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Martinelli, Giorgio (Department of Neurology; Mount Sinai School of Medicine) Friedrich, Victor (Fishberg Center for Neurobiology; Mount Sinai School of Medicine)		
Grant/Contract No.:	NCC2-1173		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this project is to identify the morphologic alterations in rat cerebellar cortex that result from transient exposure to an altered gravitational environment. The cerebellum is the part of the brain that is critical for maintaining balance and for processing information from the limbs so that they move in a coordinated fashion. Two specific aims are being pursued. The first aim is to compare the nature and extent of morphologic changes present in the cerebellar cortex of perfusion-fixed flight and ground control rats. The second aim is to quantify parameters of cerebellar morphology in rats following brief exposure to hypergravity. The objective of the proposed research is to identify the cellular bases for the ultrastructural alterations in Purkinje cells that result from brief exposure to an altered gravitational environment. Data from this study is critical for identifying the health- and performance-related risks associated with human spaceflight, and will establish a ground-based model system for evaluating the neurologic impact of repeated exposures to launch and landing stimulus profiles. The results of these studies will provide new information concerning the gravitational conditions that trigger neuronal structural re-organization in the cerebellum.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Data from this study is critical for identifying the health- and performance-related risks associated with human spaceflight, and will establish a ground-based model system for evaluating the neurologic impact of repeated exposures to launch and landing stimulus profiles.
Task Progress:	The present proposal derives from results obtained in a flight experiment on STS-90 (April 17-May 3, 1998). Cerebellar tissue was obtained from flight animals (24 adult male Fisher 344 rats) on flight day (FD) 2, FD14, return day (R+) 1 and R+13. Cerebella from equal numbers of vivarium control and cage control rats (housed in flight-type cages maintained on Earth) were also obtained (FD2 N=4; FD14 N=9; R+1 N=5; R+13 N=6). Because of the necessity for tissue sharing, as well as crew safety concerns, the animals were not sacrificed by vascular perfusion (which is preferable for ultrastructural tissue preservation). Instead, hindbrain tissue was immersion-fixed using a protocol specifically developed for the project. Tissue harvesting and processing were identical for the flight and control specimens. After fixation, each cerebellum was cut by Vibratome into 100 µm thick parasagittal sections. These were collected serially, processed for electron microscopy, and embedded as tissue wafers between plastic coverslips. Specimens of the cerebellar nodulus were dissected from these wafers, thin-sectioned, and examined by transmission electron microscopy. Qualitative ultrastructural studies of the nodulus were then conducted. In the FD2 specimens, the Purkinje cell bodies showed subtle morphologic abnormalities. For example, the whorl of organelles normally distributed throughout the somal cytoplasm was clustered in a peri-nuclear annulus. The most dramatic ultrastructural alteration in the nodulus of FD2 flight rats, however, was the presence of large numbers of extensive lamellar bodies throughout the Purkinje cell dendritic tree. Normally, cisterns of smooth endoplasmic reticulum are present in Purkinje cells and serve as an identifying ultrastructural feature of the cells. In Purkinje cells of the nodulus of FD2 flight animals, large stacked lamellar bodies were present throughout the dendrites and spines of the Purkinje cells, and were often continuous with adjacent mitochondria and the nearby plasmalemma. Ultrastructural signs of filamentous degeneration were also apparent in Purkinje cell dendrites in the nodulus of FD2 flight rats. These profiles contained degenerated mitochondria, increased numbers of lysosomes and vacuoles, and multivesicular bodies. Synaptic contacts often appeared normal in the FD2 flight noduli, with well-defined clefts and post-synaptic densities, and synaptic vesicles adherent to the presynaptic membranes. However, synapse retraction was occasionally observed in the nodulus of these animals. Such contacts were characterized by retraction of the synaptic vesicles away from the region of membrane contact, leaving an electron-lucent subjunctional region between the vesicle cluster and the presynaptic membrane. These results provide striking evidence for structural alterations in the nodulus of flight rats sacrificed 24 hr post-launch. Surprisingly, the nodulus of FD14 flight rats did not demonstrate these alterations. Purkinje cells appeared normal, with the usual complement of organelles scattered from the nuclear membrane to the plasmalemma. Unembellished hypolemmal cisternae were present throughout the Purkinje cell dendritic trees. Our interpretation of these findings is that the architecture of the cerebellar molecular layer re-establishes structural stability following sustained exposure to a constant gravitational force, regardless of the absolute value of that force. This interpretation is bolstered by our observation that the nodulus of flight rats sacrificed 24 hr after the 18-day shuttle mission appeared similar to that of the FD2 samples. To further evaluate this hypothesis, rats were placed on a 1 m centrifuge which was rotated at 50 rpm (300 deg/s) to deliver 2G of centripetal acceleration to the rats at the end of the arm. The rats were held in Plexiglas boxes and were centrifuged continuously for 4 hours. At the conclusion of the centrifugation, the rats were removed from their holders, and the experiment proceeded as above. As hypothesized, lamellar bodies were observed in the Purkinje cells of these rats. At present, these findings are being confirmed and extended to identify the cellular and molecular changes in the cerebellum resulting from rapid changes in the gravitational environment.
Bibliography Type:	Description: (Last Updated: 04/26/2007)
Abstracts for Journals and Proceedings	Friedrich, V.L. Jr.,; Martinelli, G.P.; Holstein, G.R. "Transient alterations in rat Purkinje cell ultrastructure occur early in and immediately following space shuttle flight." Society for Neuroscience Abstract 29 , Nov-2003
Books/Book Chapters	Holstein, G.R.; Martinelli, G.P. "Inflight central nervous system tissue fixation for ultrastructural studies." in "The Neurolab Spacelab Mission: Neuroscience Research in Space" Ed. Buckey, J.C..; Homick, J.L. NASA SP-2003-535, 259-262 , Jul-2003
Books/Book Chapters	Holstein, G.R. ; Martinelli, G.P. "Ultrastructural observations of the rat cerebellar nodulus after short-term spaceflight" in "The Neurolab Spacelab Mission: Neuroscience Research in Space" Ed. Buckey, J.C.; and Homick, J.L. NASA SP-2003-535, 19-26 , Jun-2004
Presentation	Holstein, G.R.; Friedrich, V.L., Jr.; Martinelli, G.R. "Alterations in Neuronal Ultrastructure Associated with Spaceflight" NASA/NSBRI Workshop on Fundamental and Applied Space Biology Dec-2003
Presentation	Holstein, G.R.; Friedrich, V.L., Jr.; Martinelli, G.P. "Anatomical studies of equilibrium and balance" Grand Rounds. Neurosciences Institute, JFK Medical Center; Edison N.J. Jan-2004

Presentation	Holstein, G.R. "Alterations in Neuronal Ultrastructure Associated with Spaceflight" 18th Annual Winter Conference on Current Issues in Developmental Psychobiology Jan-2003
Presentation	Holstein, G.R. "Alterations in Neuronal Ultrastructure Associated with Spaceflight" NSBRI Education Outreach Initiative entitled “Defying Gravity – Embracing Life in Space” Feb-2004
Significant Media Coverage	Friedrich, V.L. Jr.,; Martinelli, G.P.; Holstein, G.R. "Transient alterations in rat Purkinje cell ultrastructure occur early in and immediately following space shuttle flight." Research summary for the Society for Neuroscience Annual Meeting Press Book 29 , Nov-2003